

Deep Generative Models

Chapter 2: Data Generation Problem

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Old Classification Problem

The most basic **generative model** is

Naive Bayes

which was originally developed for classification

Say we have single-channel $k \times k$ images belonging to two classes

$$y = \begin{cases} 1 & \text{cat} \\ 0 & \text{dog} \end{cases}$$

We have access to a dataset $\mathbb{D}$

$$\mathbb{D} = \left\{ (x_i, y_i) \mid x_i \in \{0, \dots, 255\}^{k \times k}, y_i \in \{0, 1\} : i = 1, \dots, n \right\}$$

*and intend to learn a **classifier***

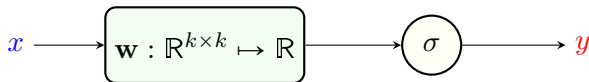
Conventional Solution: *Discriminative Modeling*

We all know the conventional solution: we consider a *discriminative* model

$$P_{\mathbf{w}}(y|x)$$

and train it on the data

In good old days, we were very *limited* in computation: it was typical to think of *simplest possible model*, i.e., a linear model



- What if we could go *even simpler*?
- + What do you mean? It's *linear*, we *cannot* think of simpler one!
- Well, we could think *tabular*!

Tabular *Discriminative* Modeling

In *tabular* approach, we estimate *distribution directly* from dataset, e.g.,

$$\hat{P}(y = 1) = \frac{\# \text{ of } y = 1 \text{'s in } \mathbb{D}}{n}$$

This approach *obviously* does not work for *discriminative* learning

+ And how is that *obvious*?

Noting that typically $n \ll 256^{k^2} \equiv \text{size of data space}$: it is *impossible* to have enough samples to get a reliable *tabular* estimate of the *posterior*

$$\hat{P}(y = 1|x) = \frac{\# \text{ of } (x, 1) \text{'s in } \mathbb{D}}{\# \text{ of samples with image } x \text{ in } \mathbb{D}}$$

Well, most of the time $\rightsquigarrow$ we are to classify *new unseen image*

both *nominator* and *denominator* are *zero*!

Tabular Generative Modeling

Unlike discriminative case, **tabular** generative learning seems possible!

In **tabular** generative learning: we need $P(x|y)P(y)$

- ① An estimate of prior belief on **label** $\rightsquigarrow$ this is easy to estimate

$$\hat{P}(y) = \frac{\# \text{ of } y\text{'s in } \mathbb{D}}{n}$$

↳ For sure, we have enough samples of **both labels**

- ② We need an estimate of the **class generative** distribution

$$\hat{P}(x|y) = \frac{\# \text{ of } (x, y)\text{'s in } \mathbb{D}}{\# \text{ of samples with label } y \text{ in } \mathbb{D}}$$

↳ For sure, the denominator is **non-zero**

↳ The nominator is however most of the time **zero!**

Tabular Generative Modeling

- + Well! We ended up with the **same issue!** Right?!
- Yes! But there is a **way out of it**

We reduce the complexity by a **simplifying assumption** $\equiv$ **tabular model**

Naive Bayes Model

In naive Bayes, we assume that **given** **y** **entries of** **x** are **mutually independent**

$$x = \begin{bmatrix} x_{11} & \dots & x_{1k} \\ \vdots & \ddots & \vdots \\ x_{k1} & \dots & x_{kk} \end{bmatrix} \longleftrightarrow P(x|y) = \prod_{i,j} P(x_{ij}|y)$$

Naive Bayes Model is Just a Naive Tabular Model

In reality, the pixels are **not** independent $\longleftrightarrow$ this is just a **simple model**

Naive Bayes Model

- + How does it help then?!
- Well, each $P(x_{ij}|y)$ is **feasible to estimate**

Say for instant we want to estimate $P(x_{11}|y)$

$$\hat{P}(x_{11} = 9|y) = \frac{\text{\# of samples in } \mathbb{D} \text{ with label } y \text{ and first entry } 9}{\text{\# of samples with label } y \text{ in } \mathbb{D}}$$

↳ With large $\mathbb{D}$, we could have **enough samples with $x_{11} = 9$**

For a new **sample x** : though we haven't seen it before, we have seen each of its pixels before $\rightsquigarrow$ We can compute its **class generative distributions as**

$$\hat{P}(x|0) = \prod_{i,j} \hat{P}(x_{ij}|0) \qquad \hat{P}(x|1) = \prod_{i,j} \hat{P}(x_{ij}|1)$$

Naive Bayes Model

To classify the new *sample* x , we can then use the *Bayes rule* and write

$$\hat{P}(y|x) = \frac{\hat{P}(x|y)\hat{P}(y)}{\hat{P}(x|0)\hat{P}(0) + \hat{P}(x|1)\hat{P}(1)}$$

and classify with the label that has higher *posterior*

Moral of Story

Sometimes *generative* modeling can offer more degrees of freedom

- ↳ It is *impossible* to think about *tabular discriminative* learning
- ↳ With *generative learning* we can use a *tabular* model

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

x_1	x_2	y
A	€	1
A	€	1
B	€	1
B	\$	0
B	\$	0
A	€	0
A	\$	0
B	€	0
B	€	1
B	\$	1

Discriminative Learning

Say, we want to classify a new sample $x = [A, \$]$

$$\hat{P}(1|x) = \frac{\#(x, 1) \text{ occurred}}{\#x \text{ occurred}} = \frac{0}{1} = 0$$

$$\rightsquigarrow \hat{P}(0|x) = 1$$

which is not *reliable* with only *one training sample*!

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

	x_1	x_2	y
→	A	€	1
→	A	€	1
	B	€	1
	B	\$	0
	B	\$	0
	A	€	0
	A	\$	0
	B	€	0
	B	€	1
	B	\$	1

Generative Learning

What if we use *naive Bayes* for $x = [A, \$]$?

$$\hat{P}(A | 1) = \frac{\#(A, 1) \text{ occurred}}{\# 1 \text{ occurred}} = \frac{2}{5} = 0.4$$

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

x_1	x_2	y
A	€	1
A	€	1
B	€	1
B	\$	0
B	\$	0
A	€	0
A	\$	0
B	€	0
B	€	1
B	\$	1



Generative Learning

What if we use *naive Bayes* for $x = [A, \$]$?

$$\hat{P}(A | 0) = \frac{\#(A, 0) \text{ occurred}}{\# 0 \text{ occurred}} = \frac{2}{5} = 0.4$$

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

x_1	x_2	y
A	€	1
A	€	1
B	€	1
B	\$	0
B	\$	0
A	€	0
A	\$	0
B	€	0
B	€	1
B	\$	1



Generative Learning

What if we use *naive Bayes* for $x = [A, \$]$?

$$\hat{P}(\$|1) = \frac{\#(\$, 1) \text{ occurred}}{\# 1 \text{ occurred}} = \frac{1}{5} = 0.2$$

So we can say

$$\hat{P}(x|1) = \hat{P}(A|1)\hat{P}(\$|1) = 0.08$$

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

x_1	x_2	y
A	€	1
A	€	1
B	€	1
B	\$	0
B	\$	0
A	€	0
A	\$	0
B	€	0
B	€	1
B	\$	1

Generative Learning

What if we use *naive Bayes* for $x = [A, \$]$?

$$\hat{P}(\$|0) = \frac{\#(\$, 0) \text{ occurred}}{\# 0 \text{ occurred}} = \frac{3}{5} = 0.6$$

So we can say

$$\hat{P}(x|0) = \hat{P}(A|0)\hat{P}(\$|0) = 0.24$$

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

x_1	x_2	y
A	€	1
A	€	1
B	€	1
B	\$	0
B	\$	0
A	€	0
A	\$	0
B	€	0
B	€	1
B	\$	1

Generative Learning

Prior probability is also readily estimated

$$\hat{P}(1) = \frac{\# \text{ 1 occurred}}{10} = \frac{5}{10} = 0.5$$

So, we have

$$\hat{P}(0) = 1 - \hat{P}(1) = 0.5$$

Example: Bayes vs Discriminative

In the old problem, assume we have only two pixels

$$x_1 \in \{A, B\}$$

$$x_2 \in \{\text{€}, \$\}$$

x_1	x_2	y
A	€	1
A	€	1
B	€	1
B	\$	0
B	\$	0
A	€	0
A	\$	0
B	€	0
B	€	1
B	\$	1

Generative Learning

What if we use *naive Bayes* for $x = [A, \$]$?

$$\begin{aligned}\hat{P}(1|x) &= \frac{\hat{P}(x|1)\hat{P}(1)}{\hat{P}(x|1)\hat{P}(1) + \hat{P}(x|0)\hat{P}(0)} \\ &= \frac{0.04}{0.16} = 0.25 \rightsquigarrow \hat{P}(0|x) = 0.75\end{aligned}$$

Sounds like *more reliable!*

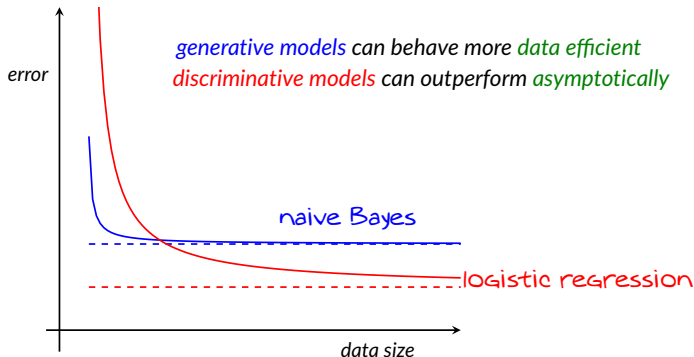
Interesting Old Study



On Discriminative vs. Generative classifiers: A comparison of logistic regression and naive Bayes

Andrew Y. Ng
Computer Science Division
University of California, Berkeley
Berkeley, CA 94720

Michael I. Jordan
C.S. Div. & Dept. of Stat.
University of California, Berkeley
Berkeley, CA 94720



Deep Generative Models

Using naive Bayes model, we can sample **new data points**

we sample $y \sim \hat{P}(y) \rightsquigarrow$ fixing y we sample $x \sim \hat{P}(x|y)$

Nevertheless, we obviously **cannot** expect **meaningful samples**

↳ the **tabular** model is **too simple**

↳ **data distribution** is **too complex**

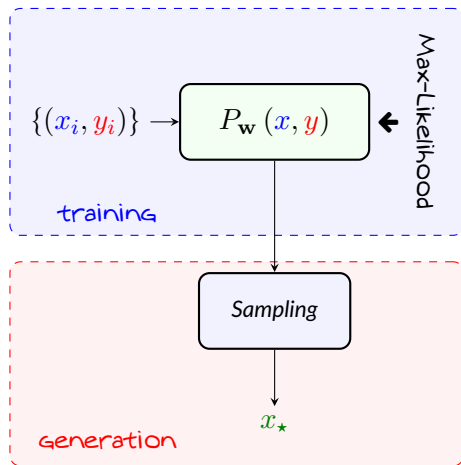
The solution is hence to go for a **deep generative model**

we sample $y \sim \hat{P}(y) \rightsquigarrow$ fixing y we sample $x \sim \hat{P}_{\mathbf{w}}(x|y)$

where $\hat{P}_{\mathbf{w}}(x|y)$ is **a deep model**, e.g., **a deep NN**

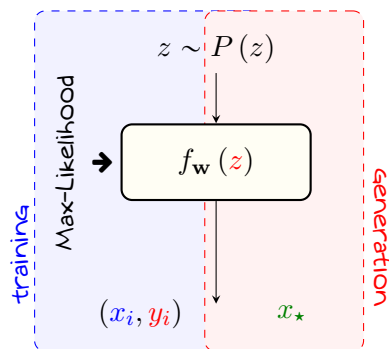
Road Map: Deep Generative Models

Conventional Deep Generation



➡ AR, Energy, Flow $\rightsquigarrow$ Chapter 3

Modern Deep Generation



➡ GAN, VAE, Diffusion

$\rightsquigarrow$ Chapters 4-6